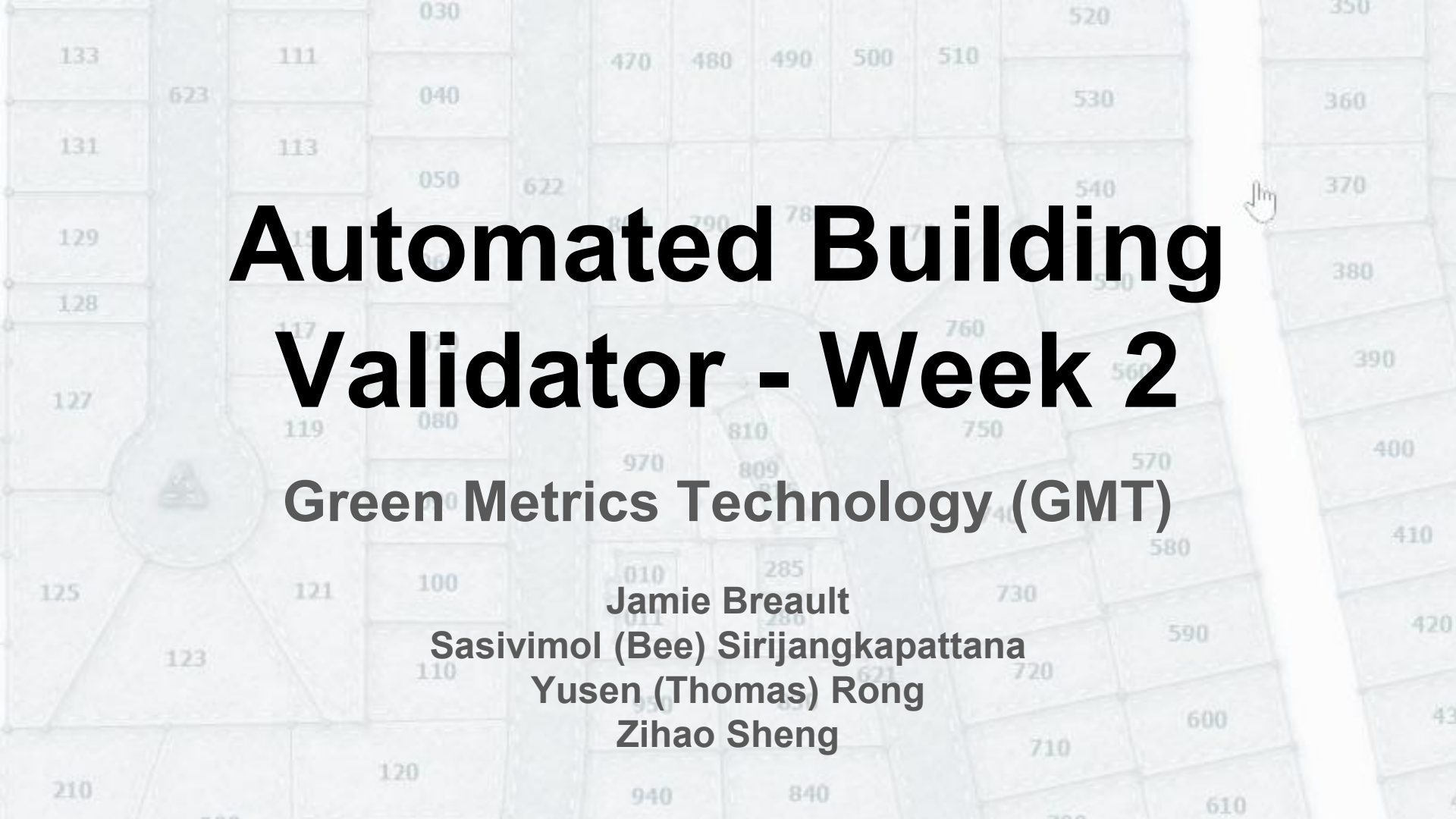


A background image of a grayscale aerial map showing building footprints and parcel boundaries. Numerous numerical parcel identifiers are visible across the map, such as 133, 111, 030, 470, 480, 490, 500, 510, 520, 623, 040, 530, 350, 131, 113, 050, 622, 540, 370, 129, 128, 127, 117, 760, 380, 119, 080, 810, 750, 560, 390, 400, 970, 809, 570, 410, 125, 121, 100, 010, 285, 730, 580, 420, 123, 110, 720, 600, 430, 210, 120, 940, 840, 710, 610, and 400. A small hand cursor icon is positioned near the top right of the map.

Automated Building Validator - Week 2

Green Metrics Technology (GMT)

Jamie Breault

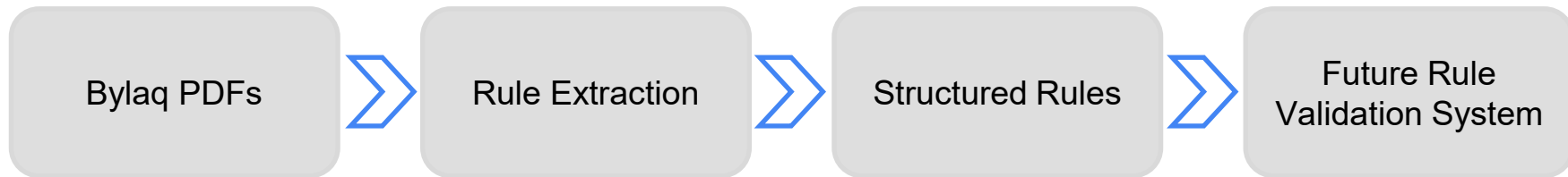
Sasivimol (Bee) Sirijangkapattana

Yusen (Thomas) Rong

Zihao Sheng

Project background

- Municipal bylaws are complex and vary across cities
- Laneway home regulations involve many zoning and building constraints
- Rules are written in different formats and structures
- Manual comparison and validation of rules is difficult and time-consuming



Updated Project Direction

- After discussions with GMT, the project scope was refined
- Initial direction
 - Broad zoning and bylaw validation system
- Current focus:
 - Laneway home rules
 - Vancouver, Surrey, Burnaby
 - Rule standardization and structure design
- Goal is to first deeply understand zoning rules before implementation
 - Understanding rule variability first
 - Building a reusable schema foundation
 - Exploring edge cases before implementation

Progress Update

- Review municipal bylaw PDFs
- Identify laneway-home-related rules
- Extract important constraints
- Standardize rules into structured format
- Compare similarities and differences across cities
- Discuss and refine reusable rule schema

Relevant Rules				
	Vancouver	Surrey	Burnaby	Rule Definitions
Definition of Single-Laneway Homes If Permitted	A detached dwelling unit constructed in the rear yard of a site on which is situated a Single Detached House, Single Detached House with Secondary Suite, or Child Day Care Facility, but does not include Infill Single Detached House.	No such concept mentioned in the bylaw, closest object could be Coach House	No such concept mentioned in bylaw, closest object could be Rear-Principle Building	
	TRUE	TRUE	TRUE	
1	max_lot_coverage	max_lot_coverage	max_lot_coverage	max_lot_coverage
2	min_front_setback	min_front_setback	min_front_setback	min_front_setback
3	min_side_setback	min_side_setback	min_side_setback	min_side_setback
4	min_back_setback	min_back_setback (called min_rear_setback)	min_back_setback	min_back_setback
5	max_height_m	max_height_m	max_height_m	max_height_m
6	max_height_stories	max_height_stories	max_height_stories	max_height_stories
7	min_outdoor_space_width	N/A for this city R1 region	min_outdoor_space_width	min_outdoor_space_width
8	min_outdoor_space_area	N/A	max_outdoor_space_area_per_unit	max_outdoor_space_area_per_unit
9	max_unit_distance_no_sprinkler	N/A for this city R1 region	max_unit_distance_no_sprinkler	max_unit_distance_no_sprinkler
10	max_fsr	* see rule 23-25, no exact fsr	N/A for this city R1 region	max_fsr
11	max_window_glazing_area	N/A for this city R1 region	N/A for this city R1 region	max_window_glazing_area
12	is_heritage (https://vancouver.ca/home-property-development/vancouver-heritage-register-vhr.aspx)	is_heritage (https://www.surrey.ca/city-government/bylaws/heritage-bylaws?utm_source=chatgpt.com)	is_heritage	is_heritage
13	N/A for this city	required_extra_parking_spaces	required_extra_parking_spaces	required_extra_parking_spaces
14	N/A for this city	frequent_bus_stop_parking_exemption	N/A for this city R1 region	frequent_bus_stop_parking_exemption
15	maximum number of dwelling units	ssmuh_max_units	ssmuh_max_units	ssmuh_max_units
16	within_400m_frequent_bus_stop	within_400m_frequent_bus_stop	Similar concept (Frequent Transit Network / Frequent Transit Network Area)	
17	N/A for this city	require_lane_access_small_lots	N/A for this city R1 region	require_lane_access_small_lots
18	allow_driveway_access_large_lots (called rear vehicle access)	allow_driveway_access_large_lots	N/A for this city R1 region	allow_driveway_access_large_lots
19	main_street_side_setback	main_street_side_setback	main_street_side_setback (called flanking/side setback)	main_street_side_setback
20	floorspace_area_limit	floorspace_area_limit	No specific rule	floorspace_area_limit
21	min_front_principal_separation (under the multiple dwelling rule)	N/A for this city R1 region	min_front_principal_separation	min_front_principal_separation
22	min_back_principal_separation (under the multiple dwelling rule)	N/A for this city R1 region	min_back_principal_separation	min_back_principal_separation
23	min_front_back_principal_separation	min_front_back_principal_separation	min_front_back_principal_separation	min_front_back_principal_separation
24	min_other_separation	min_other_separation	min_other_separation	min_other_separation
25	parking_requirements	parking_requirements (check Part 5/ table 5.1)	parking_requirements	parking_requirements
26	N/A for this city	N/A for this city R1 region	max_below_grade_garage_vertical_height	max_below_grade_garage_vertical_height
27	N/A for this city	N/A for this city R1 region	min_below_grade_garage_lot_distance	min_below_grade_garage_lot_distance
28	tree_retention	tree_retention	N/A for this city R1 region	tree_retention

Example Rule Structure

municipality	Burnaby
zone	R1
source	https://www.burnaby.ca/our-city/bylaws/zoning-bylaw
fidelity	High
key	max_lot_coverage
description	Maximum percentage of lot area that buildings can occupy
code	101.4
type	Max
value	30
condition	lot area greater than 567 m2, 1-2 units, no impervious surface
exception	null
penalty	Illegal
nullify_code	null

Challenges

- Municipal bylaws differ significantly
- Some rules are ambiguous
- Difficult to standardize all rule types
- Current work is being assigned incrementally week-by-week by GMT, so the detailed plan for next week is still evolving

Updated Next Steps

- Clarify final deliverables and expected project scope
- Discuss and refine the shared rule structure of laneway home with GMT
- Further investigate multi-building / multiple-home separation rules and edge cases

Client Communications

- Weekly Monday meeting with GMT
- Daily updates via Discord
- Blocker reports
 - Team discussion → blocker report submission → GMT feedback and clarification

Team Contributions

Team Member	Primary Tasks	Hours
Jamie	1. Coordinated communication with GMT through blocker reports 2. Led schema and rule structure discussions 3. Researched and organized Burnaby zoning bylaws and examples	37
Bee	1. Researched Vancouver R1-1 laneway home bylaws 2. Extracted and standardized zoning constraints into structured tables	32

Team Member	Primary Tasks	Hours
Thomas	1. Researched Surrey coach house / laneway home bylaws 2. Improved schema consistency and rule organization	32
Zihao	1. Quickly onboarded and caught up with the project workflow 2. Helped design rule-checking and validation workflows	32